Md Najmul Alam Patwary

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Education

Dong-Eui University **September 2025 – Ongoing**
MSc in Software and AI *Busan, South Korea*

Major Topics: Machine Learning, Deep Learning, Big Data, Embedded Systems, Human-Machine Interface, Digital Image Processing

Thesis Topic (Ongoing): UCAT-IL: Uncertainty-Aware Contact and Topology-Guided Imitation Learning from Monocular Human Demonstrations for Deformable Object Manipulation

Summary: UCAT-IL is the working name of my MSc thesis: Uncertainty-Aware Contact and Topology-Guided Imitation Learning from Monocular Human Demonstrations for Deformable Object Manipulation. The goal is to let a robot pick up a manipulation skill by watching a person perform it on ordinary video, and to cope with objects such as rope and cloth that change shape as they are handled.

Bangladesh University of Engineering and Technology **2011 – 2016**
Bachelor of Science in Mechanical Engineering *Dhaka, Bangladesh*

Thesis Topic: Flow Analysis of Laminar Wall Jet

Summary: This thesis has been carried out to find a solution of similarity to the turbulent mixing layer in the plane. A wall jet forms when a fluid leaves an orifice and flows along a surface, exhibiting both boundary layer and free-jet characteristics. The study of wall jets, particularly their velocity distribution, recirculation, and heat transfer properties, is essential to optimize their performance in practical applications. Considering the above situation, the continuity equation and the momentum equation have been derived. Then, considering the momentum equation for turbulent fluid flow, a third-order ordinary differential equation has been derived using similarity transformations which are the governing equation. Finally, a numerical solution of the governing equation has been achieved by improvisation of known boundary conditions into the initial boundary conditions. Here, MATLAB has been used to develop a computer program to solve the governing equation using the fourth-order Runge-Kutta method.

Experience

Abul Khair Group **November 2020 – August 2025**
Assistant Manager (November 2020 - August 2025)

- Design, develop, implement automation solutions using Siemens S7-300 PLC by replacing FANUC automation system in stone polishing machine.
- Created 3D models, assemblies, and detailed drawings for faucet.
- Conceptualized and created new innovative products
- Collaborate with cross-functional teams to accomplish OEE improvement project requirements and objectives, ensuring successful project execution
- Ensure compliance with electrical safety standards and best practices across the production floor, enhancing workplace safety.
- Provide technical support and training to maintenance staff and machine operators, ensuring seamless troubleshooting and operation of automated systems.
- Managed budgets related to equipment purchases and procurement, ensuring cost-effectiveness and efficiency.
- Prepared PID drawings for compressed air and cooling water line.

VitaCan Industries Ltd. **September 2017 – November 2020**
Deputy Manager (September 2019 - November 2020)

- Collaborated closely with internal and external organizations to implement LPG station, deodorization station, LPG vaporizer & aerosol "tin can" filling line along with honeywell centralize safety system from initial concept to full-scale production, including tear-down and assembly.
- Leading the mechanical design, preparing BOM, and responsible for changes to ladder logic and software including Siemens PLC, WinCC HMI, aiming to enhance Overall Equipment Effectiveness (OEE) in "aerosol can" dome and cone assembly line.
- Diagnose and resolve electrical, pneumatic, and hydraulic issues, minimizing downtime and ensuring equipment reliability.

- Developed and executed advanced proactive maintenance strategies to enhance asset reliability and reduce downtime.
- Organizing resources and material for the running project. Coordinating engineers and contractors for their particular task.
- Provide technical support and training to maintenance staff and machine operator, ensuring seamless troubleshooting and operation of automated systems.
- Managed budgets related to equipment purchases and procurement, ensuring cost-effectiveness and efficiency.
- Prepared PID drawings for compressed air and cooling water line.

Assistant Manager (September 2017 - September 2019)

- Conduct troubleshooting, diagnostics, hardware panel wiring, vision integration, and repairs on automation systems, involving 480VAC, 120 VAC, and 24VDC systems within industrial controls environment.
- Commissioning KBA supplied metal sheet printing and coating line along with X-rite supplied lab (UV paint mixing) implementation and training all of engineers for maintaining smooth production.
- Installation and arrangement of utility (chiller, AHU, air compressor, water softener) for completing aerosol Can making line, metal sheet printing and coating, mustard oil expelling, filtering & filling line.
- Conducted electrical and mechanical troubleshooting on equipment, optimizing performance and minimizing downtime.
- Troubleshooting and diagnosing pneumatic based control system of LPG filling machine.

Golden Son Ltd. **December 2015 – August 2017**

Mechanical Engineer (December 2015 - August 2017)

- Preparing CAD & CAM files from physical motor shaft and accessories sample and start production in CNC (Computer Numerical Control) Lathe machine as per supply chain planning.
- Writing M & G Code for sample spring in the CNC Spring Machine's HMI and start production as per requirement.
- Redesigning induction motor's shaft and accessories, resulting in a 1.2 percent reduction in annual material cost.
- Commissioning a WEDM (wire electric discharge machining) machine by translating from Chinese help file in the machine HMI and which was left alone for six years.
- Preparing CAD (2D) & CAM files from sample parts for sheet metal press mold and using WEDM for cutting full set of plates after heat treatment and drilling.

Language Proficiency

English IELTS: 6.0, B2
German A1, Exam not given
Korean KIIP level 1, Exam not given

Technical Skills

Languages: Python, JavaScript, C, C++
Scientific Computing: Matlab, Numpy, Scipy, Pandas, Matplotlib, PyTorch
CAD: SolidWorks, Autocad, OnShape
CFD: OpenFoam, ANSYS
Tools: Git, Github, Docker, Matlab
OS: ROS 2 (Jazzy Jalisco, MuJoCo), Linux (Ubuntu)
WORKMANSHIP: Soldering, Welding(arc)

Projects

Event-Driven Agricultural Supply Chain Coordinator Using Change Data **Ongoing**

Motivating Problem or Research Gap: Agricultural supply chains in South Asia (e.g., smallholder vegetable cooperatives in Nepal, rice/ tomato aggregators in Bangladesh, smallholder tea farmers in Sri Lanka, produce mandis in Pakistan) involve fragmented participants: farmers in SMS/USSD, rural collection centers using legacy desktop databases, cold-storage operators, transport fleets and micro-lenders. Multi-step transactions (weighing produce → quality grading → cold storage reservation → mobile money payout) frequently fail mid-flight due to power cuts, AP— timeouts, or human delays, leaving databases out of sync and farmers unpaid.

EKF Implementation in ROS 2 Environment **January 2026**

I integrated the MPU9250 IMU sensor with Raspberry Pi 5 using I^2C communication and developed ROS 2 Jazzy nodes in Python to acquire and publish real-time accelerometer, gyroscope, and magnetometer data. I implemented a quaternion-based Extended Kalman Filter (EKF) prototype incorporating gyroscope-based orientation prediction, covariance propagation, and accelerometer-based correction. I also established remote communication between Raspberry Pi and Foxglove Studio on macOS to visualize live sensor data.

Desk Study Partner with Concentration Mapper by Pomodoro Timer and Camera **September 2025**

We built a desktop study partner to maintain concentration and prepare an analysis report based on study time and eye movement by using a camera and PIR motion sensor integrated with Raspberry pi 5.

Replacing a FANUC controller with a Siemens S7-300 PLC on a stone polishing machine **January 2023**

At Abul Khair Group, I designed, developed and implemented the automation for a stone polishing machine, replacing its FANUC control system with a Siemens S7-300 PLC.

Designing a Sheet Metal Press Mold **March 2017**

We designed the sheet metal mold in Solidworks and then cut pre-required holes in the D3 tool steel metal plate for WEDM wire feeding. After performing the heat treatment process, we cut those plates in WEDM.

Designing a Solar energy Collector for Boiler feed water in industrial use **January 2016**

In this project, we have designed an open water reservoir that will be used to preheat the water and that will be used as feed water for the boiler after chemical treatment.

Thermal pile heat transfer analysis in ANSYS. **November 2015**

In this report, the analysis of the heat transfer effect in the cylindrical tube, placed underground, is done theoretically and by simulation with the help of ANSYS 16.2. Theoretical calculation is done using the Nussle number to find the overall heat transfer coefficient. Then it is used to determine the amount of heat transfer from the cylindrical pipe. The simulation result was then compared with the theoretical results to justify the analysis result.

Designing a complete working UAV using SolidWorks **June 2014**

In this case, the main wing and aileron designed using an SD7037 profile and the final wing span and fuselage length was 1091 mm and 1524 mm consecutively.

Air velocity measuring device **June 2013**

An anemometer can measure air velocity within a certain range. We reconstruct this anemometer using the features of the microcontroller ADC (analogue-to-digital converter), after finding a relation between air velocity and voltage generated by rotating an aluminum pipe mounted bucket, then the air velocity is displayed on the LCD.

Training

Programming Hero **2024 – 2025**

Trainee

- Web Development by using MERN Stack, i.e., MongoDB, ExpressJS, ReactJS, and NodeJS

Bangladesh Automation Technologies Limited **2024 – 2025**

Trainee

- Industrial Controls & Instrumentation
- Represented my class in discussions related to the welfare of the students

Polash Urea Fertilizer Factory Ltd. (PUFFL) **2015**

Trainee

- Complete Overview of 8 MW gas turbine engine with water treatment plant, Boiler, Chiller, Heat Exchanger, Multi stage compressor, Ammonia plant, Urea plant and Cooling tower.
- Learning the maintenance management of the plant.